Object detection with YOLO

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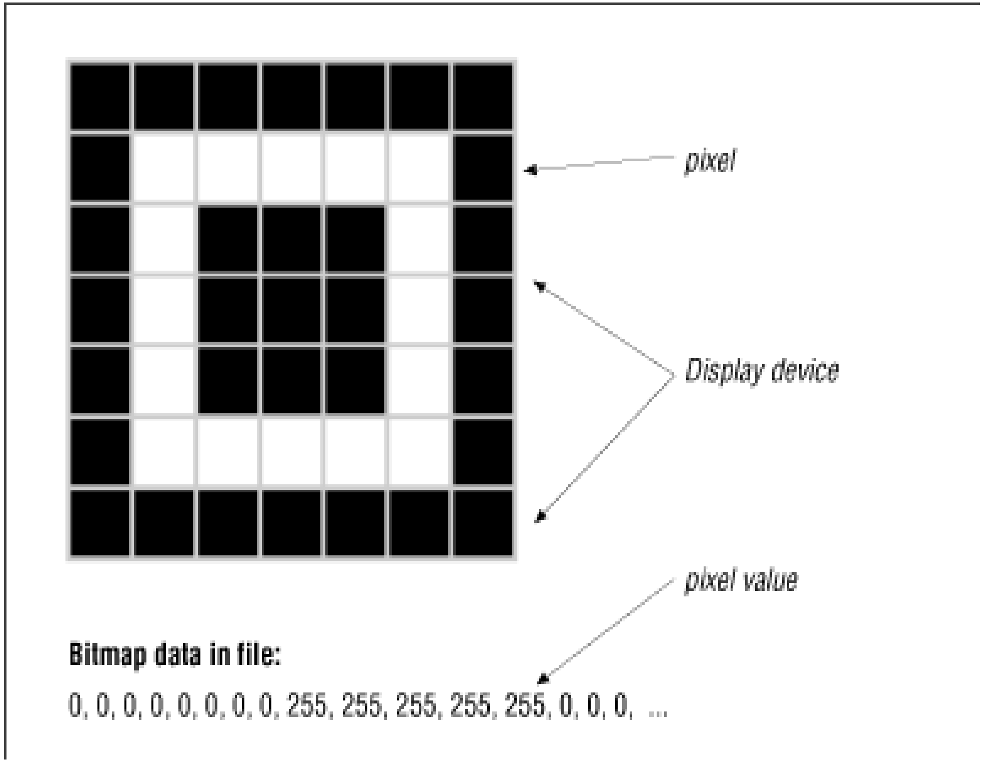
วีดีโอต้องชม

- Amazon Go
  - https://www.youtube.com/watch?v=NrmMk1Myrxc
- AI ชนะแพทย์ในการแข่งวินิจฉัยภาพสมอง
  - https://voicetv.co.th/watch/r19RKvPGQ
- “การเรียนรู้เชิงลึกอะไร | What is Deep Learning?”
  - https://www.youtube.com/watch?v=iJq8ILbmcLU

ความท้าทาย

- คอมพิวเตอร์ไม่ได้เห็นภาพแบบเรา แต่เห็นเป็น “ตัวเลข”



Link

- CS232 – Deep learning for computer vision (Stanford)
  - https://cs231n.github.io/convolutional-networks/
- Poloclub - CNN Explainer
  - https://poloclub.github.io/cnn-explainer/
- YOLO Architecture
  - https://www.geeksforgeeks.org/machine-learning/yolo-you-only-look-once-real-time-object-detection/
- Tensorflow playground https://playground.tensorflow.org/

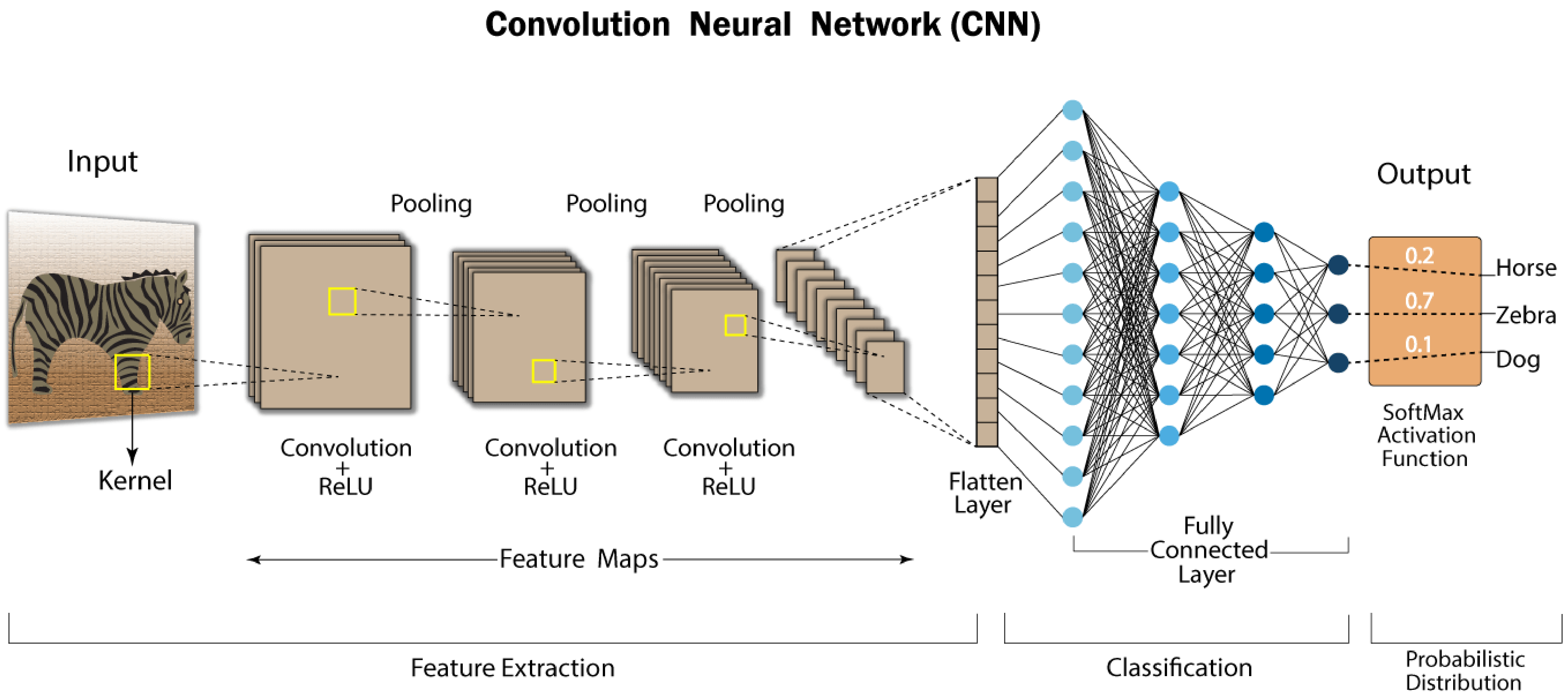


Image source: https://developersbreach.com/convolution-neural-network-deep-learning/

Methodology

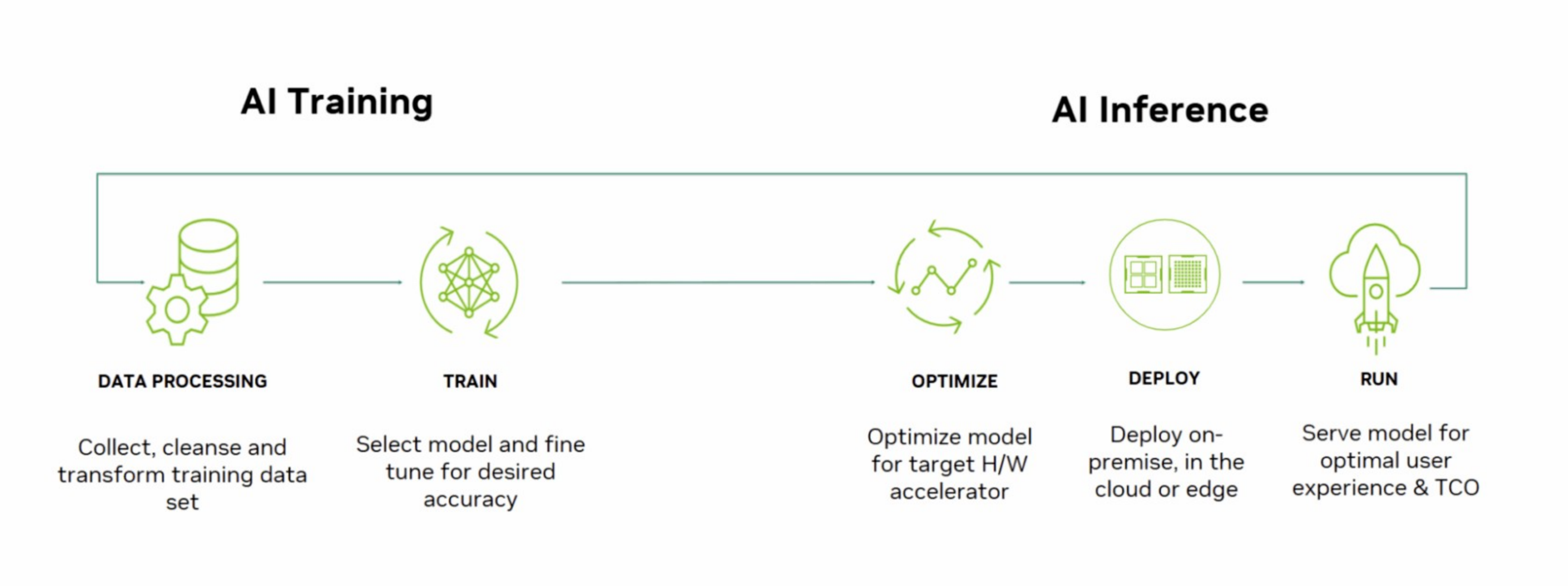


Image source: https://blogs.nvidia.com/blog/whats-difference-artificial-intelligence-machine-learning-deep-learning-ai/

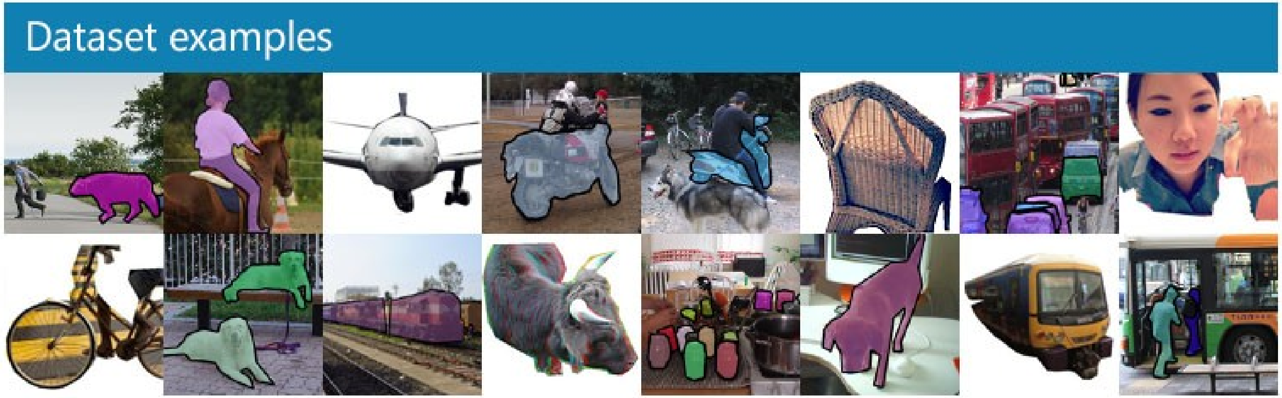
What is COCO?

COCO is a large-scale object detection, segmentation, and captioning dataset. COCO has several features:

- Object segmentation
- Recognition in context
- Superpixel stuff segmentation
- 330K images (>200K labeled)
- 1.5 million object instances
- 80 object categories
- 91 stuff categories
- 5 captions per image
- 250,000 people with keypoints

Common Objects in Context(COCO)

- It is designed to encourage research on a wide variety of object categories and is commonly used for benchmarking computer vision models.
- https://cocodataset.org/



YOLO Pre-trained Models

- YOLO - You Only Live Once
- YOLO เป็นโมเดลสำเร็จรูป ของ ultralytics ใช้ทำ
  - object detection
  - instance segmentation
  - image classification
  - pose estimation
  - multi-object tracking
- https://docs.ultralytics.com/

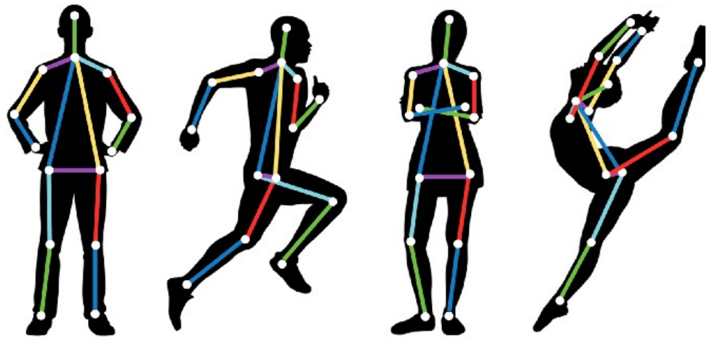


Image from: https://kemtai.com/blog/the-complete-guide-to-human-pose-estimation/

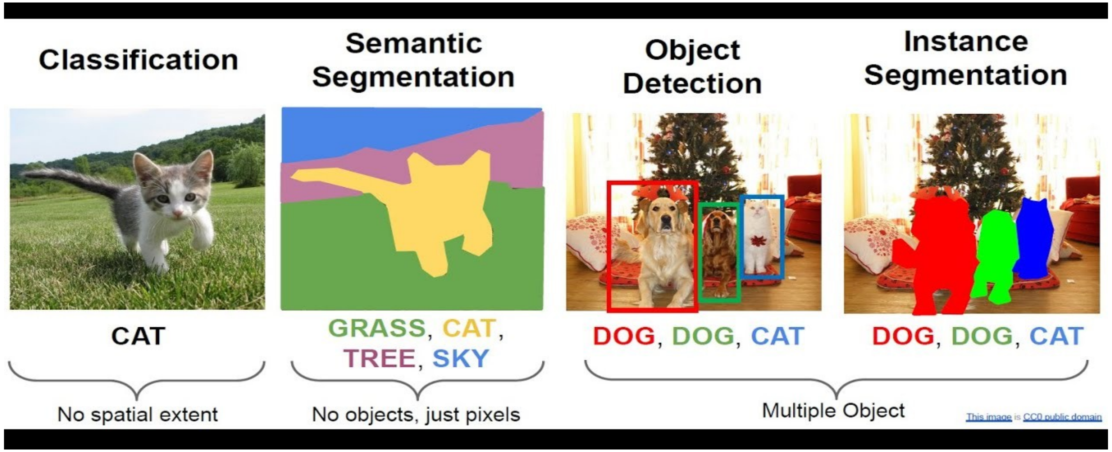


Image from: https://keymakr.com/blog/semantic-segmentation-vs-object-detection-understanding-the-differences/

Detect !

- Install

```
pip install ultralytics
```

- Run

```
from ultralytics import YOLO
import cv2

model = YOLO('yolov8n.pt')
results = model('image.jpg')
for r in results:
    im_array = r.plot()
    cv2.imshow('Object Detection', im_array)
    cv2.waitKey(0)
```

Workshop

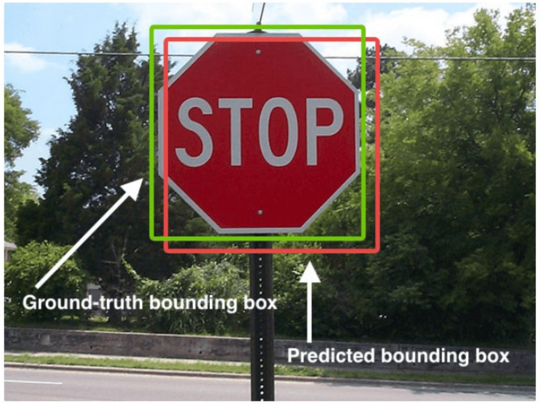
- Train model object detection ด้วยข้อมูล "ไพ่" ตามลิงค์ต่อไปนี้
  - https://github.com/EdjeElectronics/TensorFlow-Object-Detection-API-Tutorial-Train-Multiple-Objects-Windows-10
- Training steps
  - Data Collection: ถ่ายภาพไพ่ในสภาวะแสงต่างๆ
  - Annotation: ตีกรอบระบุตำแหน่ง แบบ YOLO format
  - Training: สอนโมเดลด้วย 50 Epochs
  - Validation: ทดสอบความแม่นยำด้วยภาพที่ไม่เคยเห็นมาก่อน

Label image tools

- Roboflow (web base, on Cloud)
  - https://roboflow.com/
- CVAT (Computer Vision Annotation Tool)
  - https://github.com/cvat-ai/cvat?tab=readme-ov-file
- LabelImg (App)
  - pip install LabelImg
- Label Studio (web base, on premise)
  - https://labelstud.io/guide/quick_start

Explain essentail variable

- **Epoch** คือ จำนวนครั้งที่โมเดล "อ่านข้อมูลทั้งหมด" ใน Dataset ตั้งแต่รูปแรกจนถึงรูปสุดท้ายจนครบ 1 รอบ
- **mAP** (Mean Average Precision)
  - mAP50 (Mean Average Precision at IoU 0.5)
  - IoU 0.5 (Intersection over Union)
  - mAP50 คือการวัดความแม่นยำโดยใช้เกณฑ์ความทับซ้อนที่ 50%



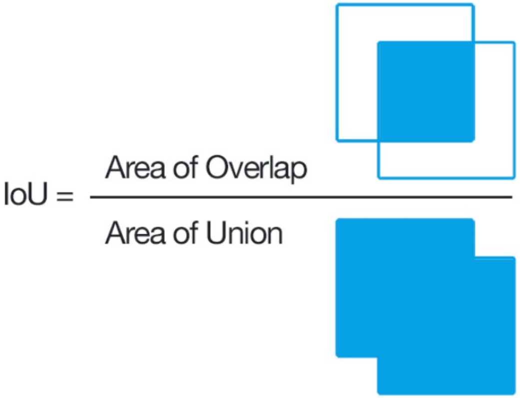
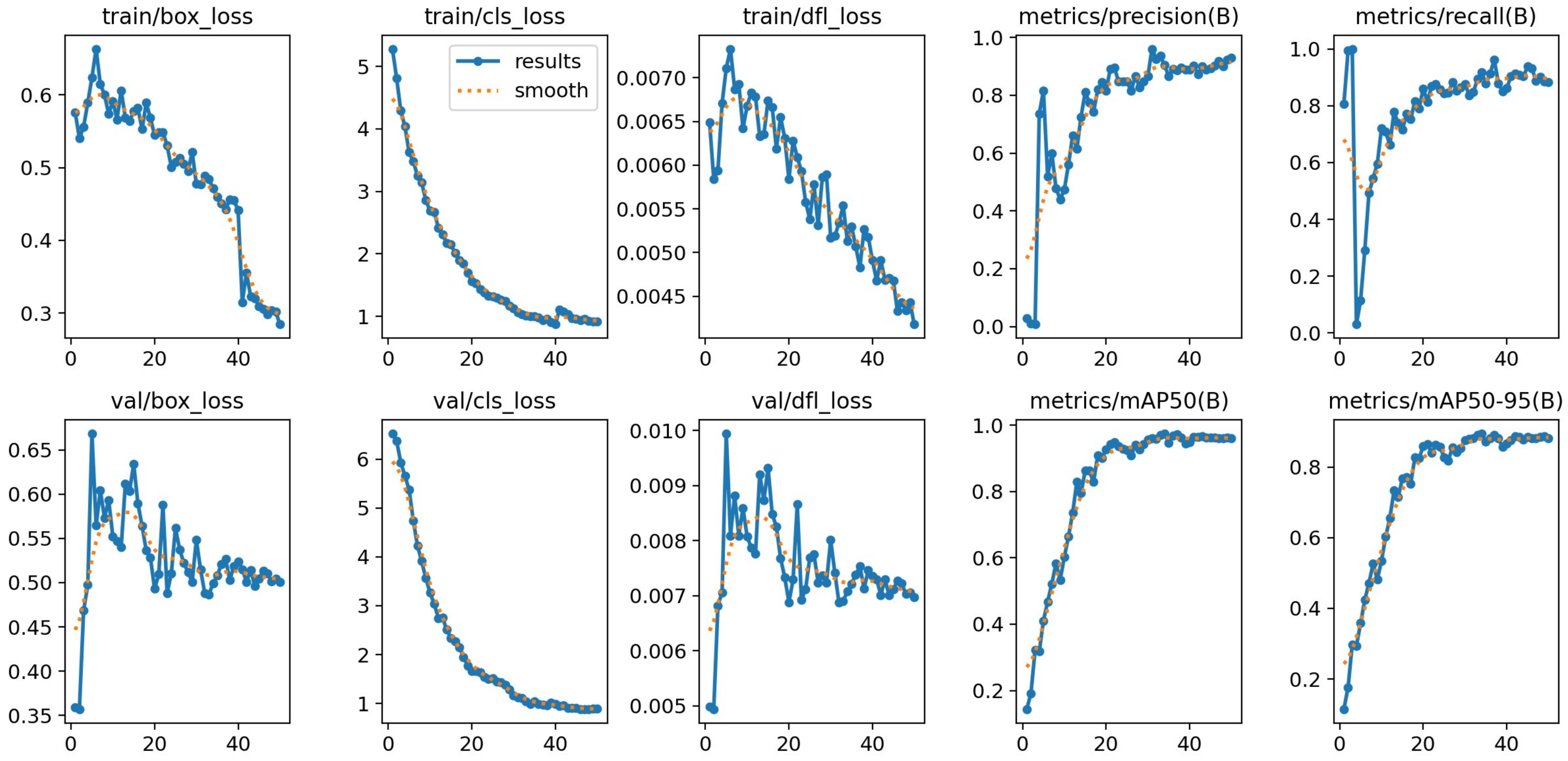


Image from: https://www.v7labs.com/blog/mean-average-precision


train/box_loss
train/cls_loss
results
smooth
train/dfl_loss
metrics/precision(B)
metrics/recall(B)
val/box_loss
val/cls_loss
val/dfl_loss
metrics/mAP50(B)
metrics/mAP50-95(B)

Source: https://github.com/developmentseed/label-maker/blob/master/examples/walkthrough-tensorflow-object-detection.md

Raster vision

Object Detection

Source: https://github.com/azavea/raster-vision-examples#how-to-run-an-example

Q & A